

1. Administrative & Study Identification

Full Study Title: A Phase 1/2, Open-Label, Dose-Escalation And Expansion Study Of Entrectinib (Rxdx-101) In Pediatrics With Locally Advanced Or Metastatic Solid Or Primary CNS Tumors And/Or Who Have No Satisfactory Treatment Options **Short Title/Acronym:** STARTRK-NG **Protocol Number:** RXDX-101-03 **NCT Number:** NCT02650401 **Posting ID:** 456921983189481822 **Sponsor/Company:** Hoffmann-La Roche / Roche **Investigational Product:** Entrectinib (RXDX-101 / Trade Name: Rozlytrek) **ATC Code:** L01EX14 **EMA URL:** [Rozlytrek EPAR Product Information](#) **Phase of Development:** Phase I/II (PHASE1, PHASE2) **Trial Status:** ACTIVE_NOT_RECRUITING **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** *Phase 1:* To determine the Maximum Tolerated Dose (MTD) and Recommended Phase 2 Dose (RP2D) of F1 and F06 formulations based on dose-limiting toxicities. *Phase 2:* To evaluate the Objective Response Rate (ORR) in patients harboring target gene fusions (NTRK1/2/3, ROS1, or ALK). **Secondary Objectives:** To evaluate duration of response (DoR), time to response (TtR), progression-free survival (PFS), overall survival (OS), and to assess the pharmacokinetics (PK, including Cmax), safety, and tolerability of entrectinib.

2.2 Background & Rationale To address the lack of satisfactory treatment options for pediatric and adolescent patients (aged ≤ 21 years) with relapsed or refractory extracranial solid tumors or primary central nervous system (CNS) tumors. This study investigates entrectinib, a potent, CNS-penetrant inhibitor of tropomyosin receptor kinase (TRK) A/B/C, ROS1, and ALK, to provide targeted therapy for patients whose tumors harbor activating fusions in these specific genes without a concomitant second oncogene.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional (Dose-escalation and basket expansion) **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A **Cohort Structure:** *Phase 1 (closed):* Relapsed/refractory extracranial solid tumors (Part A). *Phase 2:* Primary brain tumors (Part B) and extracranial solid tumors including neuroblastoma (Part D) with NTRK1/2/3 or ROS1 gene fusions. Parts C and E are closed.

3.2 Planned Enrollment & Duration Target Enrollment: 69 evaluable pediatric patients **Number of Sites:** Multicenter (US and 8 additional countries) **Estimated Study Start:** 03-May-2016 **Estimated Primary Completion:** 30-Jun-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≤ 21 years (children, adolescents, and young adults) at the time of initial study entry. 2. Histologic or molecular diagnosis of malignancy (solid tumors or primary CNS tumors) from diagnosis or relapse, with available archival tumor tissue. 3. Disease status must be measurable or evaluable per RECIST v1.1 (Parts A, C, D, E) or RANO criteria (Parts B, E), optionally using the Curie Scale where applicable. 4. Lansky or Karnofsky performance score $\geq 60\%$ and a minimum life expectancy of at least 4 weeks. 5. Recovered from acute toxic effects of prior chemotherapy, immunotherapy, or radiotherapy, with adequate organ and neurologic function. 6. Females of childbearing potential must have a negative serum pregnancy test and agree to abstinence or combined contraceptive methods. Male participants must agree to abstinence or condom use during treatment and for at least 3 months after.

4.2 Exclusion Criteria 1. Prior treatment with approved or investigational TRK or ROS1 inhibitors, or concurrent experimental therapy. 2. Receiving Enzyme Inducing Antiepileptic Drugs (EIAEDs) within 14 days of the first dose. 3. Known congenital long QT syndrome or history of recent (3 months) symptomatic congestive heart failure or ejection fraction $\leq 50\%$. 4. Familial or personal history of congenital bone disorders, bone metabolism alterations, or osteopenia. 5. Patients with neuroblastoma (NB) with bone marrow space-only disease. 6. Active gastrointestinal disease or malabsorption syndromes impacting drug absorption. 7. Known active infections or incomplete recovery from acute effects of any prior surgery. 8. Known hypersensitivity to entrectinib or its excipients.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Entrectinib administered once daily. **Route of Administration:** Oral (capsules F1 or F06 formulations given intact or as oral pellets via mouth/enteral feeding tube). **Duration of Treatment:** Continuous in 28-day (4-week) cycles until disease progression or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for managing disease symptoms. **Prohibited:** Other experimental antineoplastic therapies and Enzyme Inducing Antiepileptic Drugs (EIAEDs).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Molecular Profiling, Baseline Scans, LVEF assessment, Pregnancy Test
Baseline	Day 0	First Dose, Safety Labs, PK Sampling
Interim	Day 1 of each 28-day Cycle	Safety Labs, PK Collection, Efficacy Assessment (Tumor scans per RECIST v1.1/RANO/Curie Scale)
End of Study	Post-Treatment	Final Vital Signs, Exit Interview, IP Accountability, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Maximum Tolerated Dose (MTD)**: Assessed by National Cancer Institute Common Terminology for Adverse Events Criteria (NCI CTCAE v4.03). * **Recommended Phase 2 Dose (RP2D)**: Evaluated for F1 Formulation and F06 Formulation in pediatric participants able to swallow intact capsules, assessed by NCI CTCAE v4.03.

7.2 Secondary & Exploratory Endpoints * **Safety and Tolerability**: Adverse Events (AE), ECG, and Laboratory parameters assessed by NCI CTCAE v4.03. * **Pharmacokinetics**: Maximum observed plasma drug concentration (C_{max}) using F1 Formulation and C_{max} using F06 Formulation given intact. * **Efficacy**: Duration of Response (DoR), Progression-Free Survival (PFS), Objective Response Rate (ORR), and Overall Survival (OS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection via routine clinical assessments and laboratory tests, specifically monitoring for known TKI toxicities like weight gain, osteopenia, bone fractures, and cardiac abnormalities. Assessed via NCI CTCAE v4.03. **Serious Adverse Events (SAEs)**: Required reporting within 24 hours of investigator awareness. **Data Monitoring Committee (DMC)**: Internal and independent review mechanisms to oversee safety, dose-escalation phases, and DLTs.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients who received ≥ 1 dose of entrectinib); Efficacy-Evaluable Set (patients with target fusions, measurable baseline disease, and ≥ 1 dose at RP2D). **Sample Size Justification**: Open-label dose-escalation framework with targeted expansion baskets (Targeted Enrollment: 69), powered to establish an accurate toxicity profile and

clinical ORR for a rare pediatric population. **Handling of Missing Data:** Time-to-event efficacy variables (PFS, OS, DoR) censored at the date of the last evaluable tumor assessment.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: Conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite and remote data monitoring to verify protocol adherence. **Audit Trail:** Robust eCRF locking and Source Data Verification (SDV) coupled with central blinded independent review.

11. Study Identifiers & Terminology Mapping

Primary ID: RXDX-101-03 **Registry Number:** NCT02650401 **Posting ID:** 456921983189481822 **Sponsor/Lead Organization:** Hoffmann-La Roche / Roche **Study Title:** Study Of Entrectinib (Rxdx-101) in Children and Adolescents With Locally Advanced Or Metastatic Solid Or Primary CNS Tumors And/Or Who Have No Satisfactory Treatment Options **Official Regulatory Title:** A Phase 1/2, Open-Label, Dose-Escalation And Expansion Study Of Entrectinib (Rxdx-101) In Pediatrics With Locally Advanced Or Metastatic Solid Or Primary CNS Tumors And/Or Who Have No Satisfactory Treatment Options

12. Clinical Framework (Oncology / Specific Condition)

Disease Areas: Solid Tumors, CNS Tumors **Preferred Name:** Central Nervous System Neoplasms **Preferred UMLS Name:** Pediatric Neoplasms **Semantic Type:** Neoplastic Process **Definition:** A benign or malignant, primary or metastatic neoplasm that affects the brain, meninges, or spinal cord. Representative examples of primary neoplasms include astrocytoma, oligodendroglioma, ependymoma, and meningioma. Representative examples of metastatic neoplasms include carcinoma and leukemia. **Terminology Codes:** * **MeSH Code:** D016543 * **HPO Code:** HP:0100006 * **SNOMED ID:** 126951006 **Medical Methodology:** Targeted Tyrosine Kinase Inhibitor (TKI) Therapy

13. Study Procedures & Methodology

Management Pathway: Targeted oncology clinical pathway via molecular tumor profiling (nucleic acid-based diagnostic testing). **Education Protocol (HCP):** Protocol training on RECIST/RANO criteria, NCI CTCAE v4.03 grading, dose adjustments, and DLTs. **Education Protocol (Patient):** Counseling on oral capsule/pellet administration, contraception requirements, and AE reporting. **Quality Audit Mechanism:** Independent central review for primary efficacy validation.

14. Observation & Follow-Up Schedule

Total Duration: Up to 120 follow-up months through 2026 study completion
Onsite Visit Cadence: Day 1 of every 28-day cycle
Remote Monitoring Frequency: Survival follow-up post-treatment
Checklist Review Interval: Tumor imaging assessments regularly scheduled to evaluate RECIST/RANO/Curie Scale status

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				